

Chapter 9 :

FET Equivalent Circuits and AC Analysis

9.1 Introduction

The constructions and operation of FETs have been described in Chapter 6. The DC biasing circuits for FETs in different configurations have been discussed in Chapter 8. In this chapter, we will discuss the equivalent circuits of FETs and their performance for AC signals. Like BJT, FETs are also used as amplifiers and switches. The switching circuits using FETs have been discussed in Chapter 8. When used as amplifier, the AC signal is applied to the input terminals, and a signal with larger p-to-p voltage or current is produced at the output terminals. The input source and the output load are generally coupled connected to the amplifier via two coupling capacitors. The reasons of using these coupling capacitors have been discussed in Chapter 5 for BJT circuits. For the same reasons the coupling capacitors are used in the FET amplifiers. We can recall that these coupling capacitors are used so that the DC operating condition of the biasing circuits is not changed by the source internal resistance and the load resistance. Similarly, bypass capacitors are used to increase the AC voltage gain of the amplifiers. Because of the capacitor-coupled load, and the bypass capacitor, the AC load of the amplifier is changed and a new AC load-line, which is different from the DC load line, is produced. The AC load line for FET amplifier will be discussed here again.

Like BJT amplifiers, FET can be used as amplifier in three configurations. Already in Chapter 8, we have found their names. The amplifiers are: common source, common drain, and common gate amplifier. In this chapter, we will determine the voltage gain, input impedance, output impedance, and current gain of these amplifiers. To do that, first, we will discuss how to draw the equivalent circuit of FETs.

9.2 AC Load Line of FET Amplifier

We know that, the load line is a line which is drawn on the output characteristic curves with a slope of $-1/R_L$. Therefore, the load line depends on the load resistance of the amplifier. For the amplifier of Fig.9-1(a), the DC load line is drawn using the following equation.

$$V_{DS} = V_{DD} - I_D(R_D + R_S) \quad (9-1)$$

But, from the equivalent circuit of Fig.9-1(b), we find that for AC signal, R_S is not effective. Moreover, the AC current passes through the parallel combination of R_D and R_L . That is, the effective load for the AC signal is $R_D \parallel R_L$.

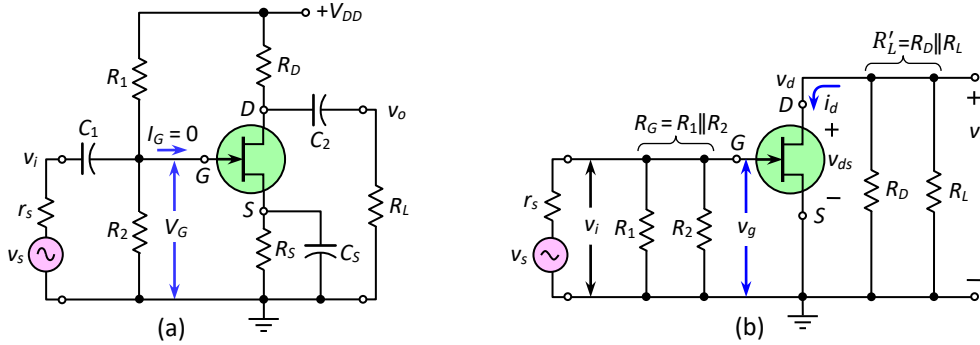


Fig.9-1: (a) JFET common source amplifier, and (b) AC equivalent circuit of this amplifier

As discussed for BJT circuit, we know that the AC load line passes through the DC operating point. Therefore, if we draw a straight line through the DC operating point with a slope of $-1/R_D \parallel R_L$, this will be the AC load line (shown in Fig.9-2). AC load line can also be drawn using a suitable change in drain current (ΔI_D) and the corresponding change in drain-source voltage (ΔV_{DS}). The process is further elaborated in Example 9-1.

When AC signal is applied to an amplifier, the operating point of the circuit oscillates along the AC load line not along the DC load line.

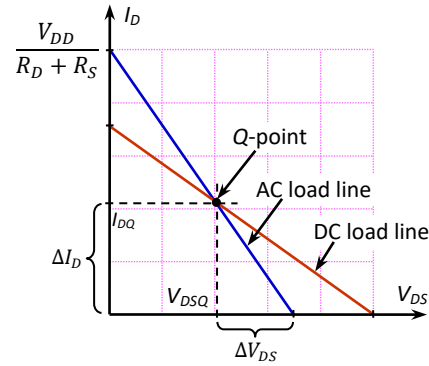


Fig.9-2: AC and DC load line of CS amplifier

Example 9-1

For the common source amplifier with voltage divider network of Fig.9-3, draw the DC and AC load line. Assume, $V_p = -5$ V, and $I_{DSS} = 10$ mA.

Solution:

The DC load line can be drawn using Equ.(9-1).

$$V_{DS} = 20 \text{ V} - I_D(1.5 \text{ k} + 1 \text{ k}) = 20 \text{ V} - 2.5 \text{ k} \times I_D$$

When $I_D = 0$, $V_{DS} = 20$ V. This point will be on the x -axis.

When $V_{DS} = 0$, $I_D = 20 \text{ V}/2.5 \text{ k} = 8$ mA. This point lies on the y -axis.

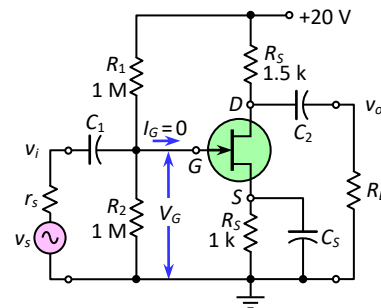


Fig.9-3: Circuit for Example 9-1

Putting these points on a $I_D - V_{DS}$ coordinate system, the load line is drawn as shown in Fig.9-4.

Determination of operating point:

The value of V_G can be calculated as,

$$V_G = \frac{V_{DD}R_2}{R_1 + R_2} = \frac{20V \times 1\text{ M}}{1\text{ M} + 4.3\text{ M}} = 3.77\text{ V}$$

From Fig.9-3 the bias line of this circuit will be,

$$V_{GS} = V_G - I_D R_S$$

Putting this value in the Shockley's equation we get,

$$I_D = I_{DSS} \left(1 - \frac{V_G - I_D R_S}{V_P} \right)^2$$

or

$$I_D = 10\text{ mA} \left(1 - \frac{I_D - 3.77}{5} \right)^2$$

Solving this quadratic equation, we get, $I_D = 5.17\text{ mA}$.

$$\therefore V_{DS} = 20\text{ V} - 5.17\text{ mA}(1.5\text{ k} + 1\text{ k}) \approx 7.1\text{ V}$$

Using these values (on the load line), the operating point is determined and shown in Fig.9-4.

Ac load of the circuit is, $R'_L = R_D \parallel R_L = 1.5\text{ k} \parallel 3\text{ k} \approx 1\text{ k}$

$$\text{Let, } \Delta I_D = 5.17\text{ mA} \quad \therefore \Delta V_{DS} = R'_L \times \Delta I_D = 1\text{ k} \times 5.17\text{ mA} \approx 5.17\text{ V}$$

Using these changes in current and voltage, the AC load line is drawn through the DC operating point. The DC load line, operating point, and the AC load line are shown in Fig.9-4.

Comments: As the AC load resistance (R'_L) is lower than DC resistance ($R_D + R_S$), the AC load line is steeper than the DC load line.

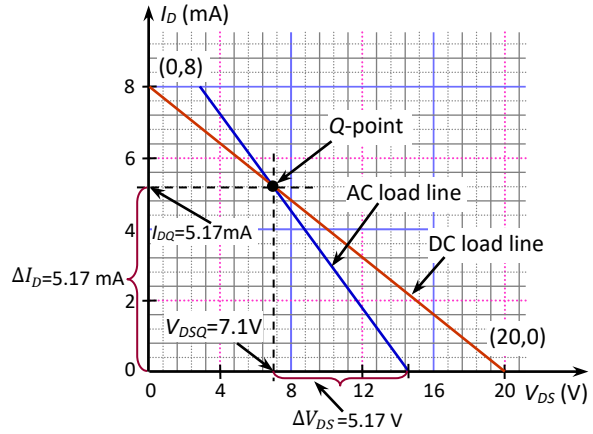


Fig.9-4: AC and DC load lines of Example 9-1

9.3 FET Equivalent Circuit

In Chapter 6, we have studied the equivalent circuit of BJT. Equivalent circuits (or transistor models) help to analyze circuits. Like BJT, a FET can be represented by an equivalent circuit. The equivalent circuit of FET is comparatively easier. The low and medium frequency equivalent circuit of a FET in common source configuration is shown in Fig.9-6. Although, here the equivalent circuit for a common source configuration is shown, this model can be used for other two configurations, with changing the position of the terminals. Here, the input signal is applied between gate-source terminal, and the output is taken from the drain-source terminal. As the gate source junction is always reverse biased, the applied signal (v_{gs}) faces a very high resistance which is represented by R_{gs} . R_{gs} is the leakage resistance, and has a high value, in the order of $10^9\ \Omega$.

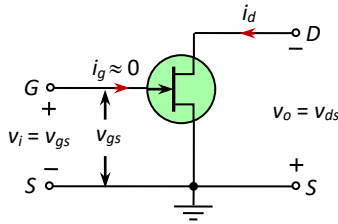


Fig.9-5: FET in CS configuration

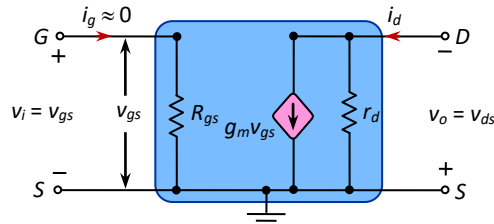


Fig.9-6: Equivalent circuit of FET in CSC

The FET produces a drain current, proportional to v_{gs} , at the output terminal. Therefore, the output stage is represented by a constant current-source. The value of the current of this source is $g_m v_{gs}$. Here, g_m is the **forward transfer admittance** or **trans-conductance** of FET, and v_{gs} is the input voltage. As discussed in Chapter 6, for small signal JFETs, the value of g_m is 1 to 30 mS.

When used as amplifier, FET works in the *pinch-off region* (or active region). As shown in Fig.9-7, the output characteristic curves, in the pinch-off region, are not completely parallel to x -axis rather they have a very small slope. This slope corresponds to a very large resistance r_d . As this resistance affects the AC signal, it is called **AC output resistance**. In the equivalent circuit, this resistance is connected in parallel with the constant current-source. The FET datasheets, sometimes mention the value of Y_{os} instead of r_d . From Y_{os} the value of r_d can be calculated as $r_d = 1/Y_{os}$. The typical value of r_d is 20 k Ω to 100 k Ω .

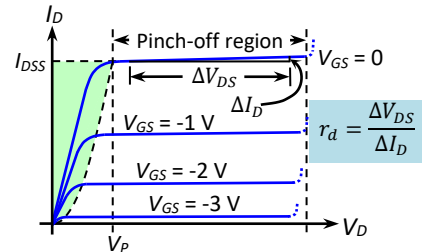


Fig.9-7: Output AC resistance of FETs

The equivalent circuit of Fig.9-6 is for FET common source configuration. By changing the positions of the terminals the equivalent circuit of other two configurations can be drawn. Fig.9-8 and Fig.9-9 show the equivalent circuit of FET in CD and CG configuration, respectively.

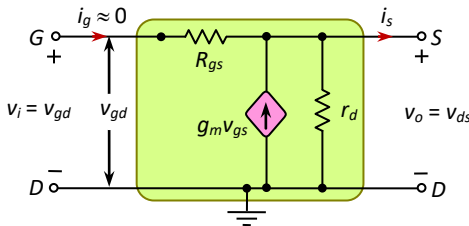


Fig.9-8: FET equivalent circuit in CD configuration

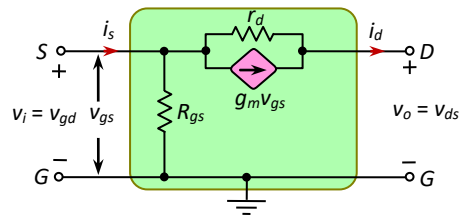


Fig.9-9: FET equivalent circuit in CG configuration

9.4 FET Common Source Amplifier

The common source (CS) amplifier is mostly used FET amplifier. Fig.9-10 shows a single stage, common source amplifier. It is also called **capacitor-coupled single stage amplifier**, as the input source and the output load are connected to the amplifier via coupling capacitors (C_1, C_2). Here,

the input signal is applied to the gate and source terminals, and the output is taken from the drain and source terminals. The source is common to both the input and output. For this reason, this amplifier is called **common source amplifier**. Common source amplifier provides high voltage gain, and high input impedance.

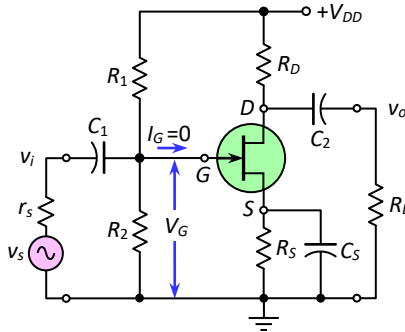


Fig.9-10: FET common source (CS) amplifier

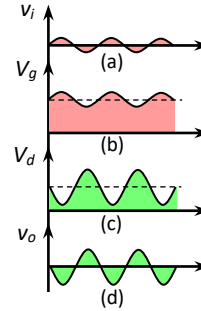


Fig.9-11: Waveforms at different points of CS amplifier

9.4.1 Operation of CS Amplifier

To amplify, the AC signal is applied in the input of the amplifier. The signals at different levels of the amplifier are shown in Fig.9-11. The input signal [Fig.9-11(a)] has positive and negative half-cycles with very small amplitude. When this signal appears to the gate through C_1 , it superimposes with the gate biasing voltage V_G [Fig.9-11(b)]. For the positive half-cycles, the gate voltage increases and consequently drain current also increases. With the increase in I_D , the voltage-drop across R_D increases and hence, the drain voltage decreases [Fig.9-11(c)]. On the other hand, for the negative half-cycles the gate voltage decreases, the drain current decreases and hence, the drain voltage increases. As the output is taken from drain, the output decreases for the increase in input and vice versa. Thus, the common source amplifier produces 180° phase difference between the input, and output signals. In addition to that, due to large drain resistance (R_D), a small change in drain current produces a larger change in drain voltage. In this way, an amplified signal is developed at the output. The output signal, developed at the drain terminal, has a DC voltage level as shown in [Fig.9-11(c)]. But, when this signal passes through the coupling capacitor (C_2), and appears at the load resistance, the DC voltage level is blocked and a pure AC output is produced as shown in [Fig.9-11(d)].

9.4.2 AC Analysis

The AC equivalent circuit of the common source amplifier is shown in Fig.9-12. The final equivalent circuit of the amplifier is drawn by replacing the FET by its equivalent model, which is shown in Fig.9-13. Here, R_G is the equivalent resistance of parallelly connected biasing resistance R_1 and R_2 . The directions of current and the polarities of voltages, shown in

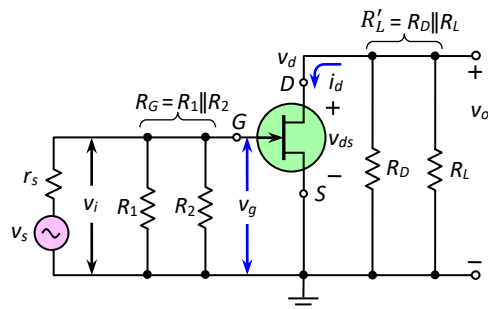


Fig.9-12: AC equivalent circuit of CS amplifier

this figure, are those for the positive half-cycles of input signal.

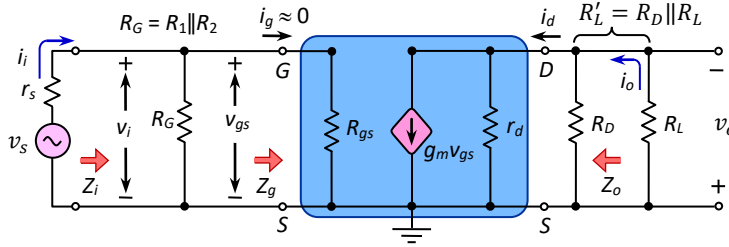


Fig.9-13: AC equivalent circuit of CS amplifier with FET model

Input Impedance:

As shown in , the value of input impedance at gate terminal will be,

$$Z_g = R_{gs} \quad (9-2)$$

When measured at the input terminal, its value will be,

$$Z_i = R_{gs} || R_G \quad (9-3)$$

As we know, the value of the leakage resistance (R_{GS}) is very large (few hundred M Ω). Therefore, the input impedance can be represented as,

$$\text{Considering } R_{GS} \approx \infty, \quad \boxed{Z_i = R_G = R_1 || R_2} \quad (9-4)$$

Output Impedance:

To measure the output impedance, the input terminal has to be shorted, i.e., $v_{gs} = 0$. Therefore, the value of the output current-source will be zero. It means that the output current-source will work as open circuit. The output impedance of the amplifier will be,

$$Z_o = R_D || r_d \quad (9-5)$$

But, as we know, the value of r_d is very large compared to R_D .

$$\because r_d \text{ is very large,} \quad \boxed{Z_o \approx R_D} \quad (9-6)$$

The impedance considering the load resistance, that is the impedance seen by the output signal, will be,

$$Z_o \approx R_D || R_L \quad (9-7)$$

Voltage Gain:

As shown in Fig.9-13, the input voltage is,

$$v_i = v_{gs} \quad (9-8)$$

At the output stage, drain current i_d flows through the parallel combination of R_D and R_L , while $g_m v_{gs}$ amount of current flows through the parallel combination of three resistors (r_d , R_D and R_L). Therefore, we can write,

$$v_o = -i_d(R_D || R_L) = -g_m v_{gs}(r_d || R_D || R_L) \quad (9-9)$$

$$\because r_d \text{ is very large,} \quad v_o = -g_m v_{gs} (R_D \parallel R_L) \quad (9-10)$$

Now, the voltage gain of the amplifier can be calculated using Equ.(9-8) and Equ.(9-10) as,

$$A_v = \frac{v_o}{v_i} = \frac{-g_m v_{gs} (R_D \parallel R_L)}{v_{gs}} = -g_m (R_D \parallel R_L) = -g_m R'_L \quad (9-11)$$

$$\text{or,} \quad \boxed{A_v = -g_m (R_D \parallel R_L) = -g_m R'_L} \quad (9-12)$$

Current Gain:

Being a voltage-controlled device, FET has an infinite current gain. But, as the biasing resistors of the amplifier circuit draws some current from the input source, the amplifier has a finite current gain. On the other hand, for small signal amplifier current gain is not important. However, to get some idea about the current gain of different FET amplifiers, here the current gain is also calculated.

As shown in Fig.9-13 the input current will be,

$$i_i = \frac{v_i}{Z_i} = \frac{v_{gs}}{R_G} \quad (9-13)$$

At the output stage, current $g_m v_{gs}$ is divided among r_d , R_D , and R_L . The current flowing through R_L is the output current. However, as the value of r_d is very large compared to the other two resistors, we can write,

Output current will be (considering r_d is very large),

$$\text{Neglecting } r_d, \quad i_o = \frac{g_m v_{gs} R_D}{R_L + R_D} \quad (9-14)$$

Now, the voltage gain of the amplifier can be calculated using Equ.(9-13) and Equ.(9-14) as,

$$A_i = \frac{i_o}{i_i} = \frac{g_m v_{gs} R_D}{R_L + R_D} \times \frac{R_G}{v_{gs}} \quad (9-15)$$

$$\boxed{A_i = \frac{g_m R_D R_G}{R_L + R_D}}$$

Example 9-2

Calculate, input impedance, output impedance, voltage gain, and current gain of the amplifier of Fig.9-14. Assume, $g_m = 6 \text{ mS}$.

Solution:

Here,

$$R_G = R_1 \parallel R_2 = 4.3 \text{ M} \parallel 1 \text{ M} \approx 811 \text{ k} \text{ [Ans.]}$$

Z_i :

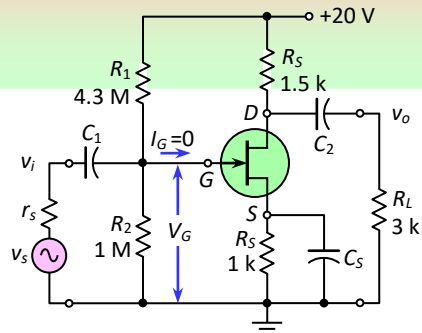


Fig.9-14: Circuit for Example 9-2

Using Equ.(9-4), $Z_i = R_1 \parallel R_2 = 4.3 \text{ M} \parallel 1 \text{ M} \approx 811 \text{ k}$ [Ans.]

Z_o :

Using Equ.(9-5), $Z_o \approx R_D = 1.5 \text{ k}$ [Ans.]

A_v :

Using Equ.(9-12), $A_v = -g_m(R_D \parallel R_L) = -6 \text{ mS} \times (1.5 \text{ k} \parallel 3 \text{ k}) = -6$ [Ans.]

A_i :

Using Equ.(9-15), $A_i = \frac{g_m R_D R_G}{R_L + R_D} = \frac{6 \times 1.5 \text{ k} \times 811 \text{ k}}{3 \text{ k} + 1.5 \text{ k}} = 1600$

Comments: FET amplifiers give more input impedance and current gain, but less voltage gain than BJT amplifiers.

9.5 Analysis of CS Amplifier with Fixed and Self-Bias

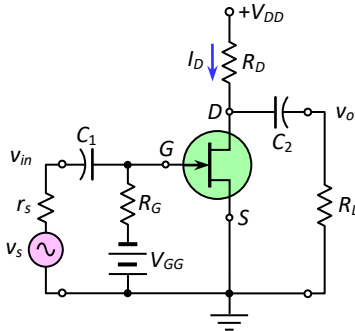


Fig.9-15: CS amplifier with fixed bias

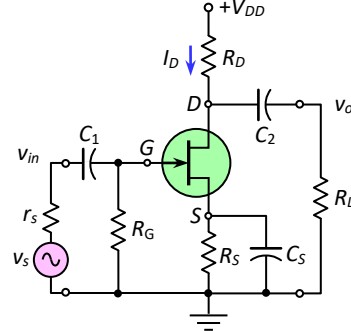


Fig.9-16: CS amplifier with self-bias

In addition to the voltage divider bias, we have discussed fixed or gate bias and self-bias circuits in Chapter 8. The biasing circuits are reproduced in Fig.9-15 and Fig.9-16.

To draw the AC equivalent of these circuits, we have to consider all capacitors and voltage sources as short circuits. The AC equivalent circuits will be same as is shown in Fig.9-17. Now, if we compare this equivalent circuit with the AC equivalent circuit with voltage divider bias [Fig.9-12], we find that they are same. Only difference is for voltage divider bias $R_G = R_1 \parallel R_2$, but in case of fixed and self-bias, R_G is a single resistance.

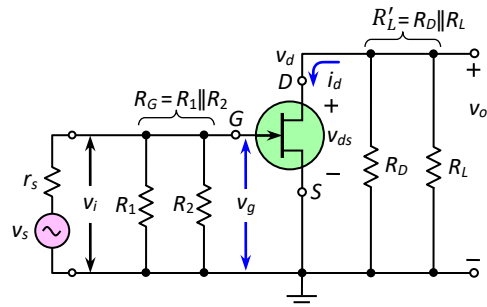


Fig.9-17: AC equivalent circuit of CS amplifier with fixed bias, and self-bias

Therefore, the expressions of input impedance, output impedance, voltage gain and current gain derived for the CS amplifier with voltage divider bias can be used for the amplifiers with fixed bias

and self-bias. In place of $R_1 \parallel R_2$, we have to use only R_G . The process is further explained in Example 9-3.

Example 9-3

For the following [Fig.9-18] common source amplifier with fixed bias, calculate input impedance, output impedance, voltage gain, and current gain. Assume $g_m = 3.5 \text{ mS}$.

Solution:

Z_i : We can use Equ.(9-4) as,

$$Z_i = R_G = 1 \text{ M}\Omega \text{ [Ans.]}$$

Z_o : Using Equ.(9-5),

$$Z_o \approx R_D = 2 \text{ k} \text{ [Ans.]}$$

A_v :

Using Equ.(9-12), $A_v = -g_m(R_D \parallel R_L) = -6 \text{ mS} \times (1.5 \text{ k} \parallel 3 \text{ k}) = -6 \text{ [Ans.]}$

If there is no R_L , $A_v = -g_m R_D = -3.5 \text{ mS} \times 2 \text{ k} = -7 \text{ [Ans.]}$

A_i :

Using Equ.(9-15), $A_i = \frac{g_m R_D R_G}{R_L + R_D} = \frac{6 \times 1.5 \text{ k} \times 1 \text{ M}}{3 \text{ k} + 1.5 \text{ k}} = 2000 \text{ [Ans.]}$

Comments: Using same process CS amplifier with self-bias can also be analyzed.

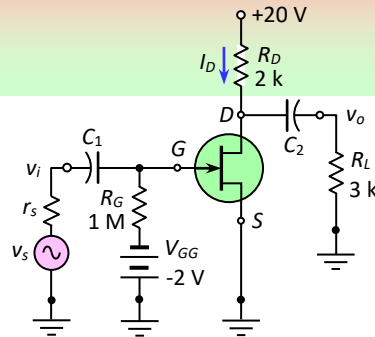


Fig.9-18: CS amplifier with fixed bias for Example 9-3

9.6 AC Analysis of Common Source Amplifier with Unbypassed Source Resistor

Fig.9-19 shows a common source FET amplifier without source bypass capacitor. If the source resistor is not bypassed, it will be effective for the AC signal. In the AC equivalent circuit of a CS

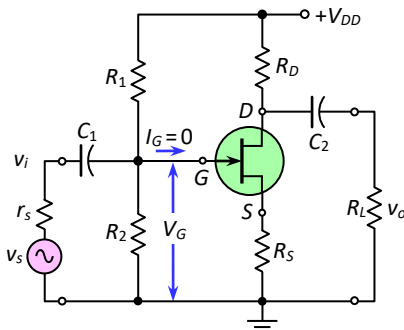


Fig.9-19: CS amplifier with unbypassed source resistor

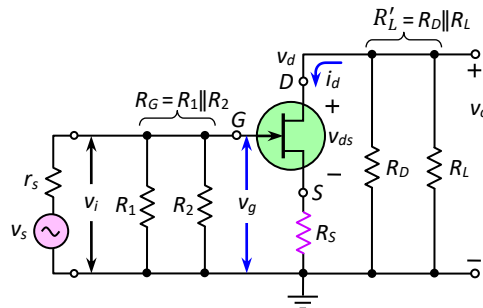


Fig.9-20: AC equivalent circuit of CS amplifier with unbypassed source resistor

amplifier with bypass capacitor [Fig.9-12], the R_S was short circuited. But now, R_S will be present in the AC equivalent circuit as shown in Fig.9-20.

Now, to analyze the circuit, we will replace the FET by its equivalent model. The final equivalent circuit is shown in Fig.9-21 and different parameters of the amplifier are calculated below.

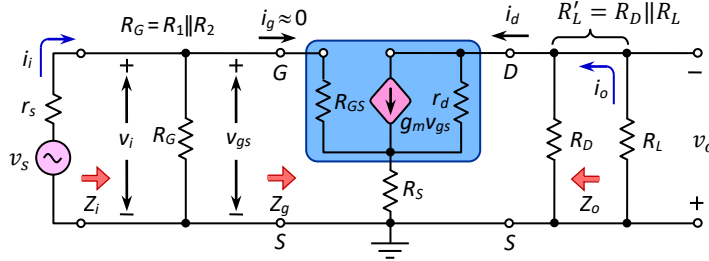


Fig.9-21: AC equivalent circuit of CS amplifier with unbypassed source resistor (using FET model)

Input Impedance:

Using KVL in the input side of the equivalent circuit, we can write the expression of input of input voltage as,

$$v_i = v_{gs} + i_d R_S = v_{gs} + g_m v_{gs} R_S = v_{gs} (1 + g_m R_S) \quad (9-16)$$

Gate circuit will be,
$$i_g = \frac{v_{gs}}{R_{GS}} \quad (9-17)$$

The impedance at gate terminal can be calculated as,

$$Z_g = \frac{v_i}{i_g} = v_{gs} (1 + g_m R_S) \times \frac{R_{GS}}{v_{gs}} = (1 + g_m R_S) R_{GS} \quad (9-18)$$

So, the impedance at the input terminal will be,

$$Z_i = R_1 || R_2 || Z_g = R_G || Z_g \quad (9-19)$$

Here, Z_g is very large compared to $R_1 || R_2$, so, the input impedance can be calculated as,

$$Z_i \approx R_1 || R_2 \quad (9-20)$$

Output Impedance:

To calculate the output impedance, we have to short the input terminals, and apply an AC signal v_o at the output as shown in Fig.9-22. Measuring i_d for this applied signal the output impedance is calculated. Now, the voltage-drop across R_S will work as v_{gs} . Therefore,

$$v_{gs} = i_d R_S \quad (9-21)$$

The v_{gs} developed in this way, produces an AC output current that opposes the current due to v_o . The value of this opposing current is,

$$i = -g_m v_{gs} = -g_m i_d R_S \quad (9-22)$$

Now, we will convert the current generator of FET model ($g_m v_{gs}$ in parallel with r_d) into a voltage generator ($g_m v_{gs} r_d$ in series with r_d). Considering R_G very large, the converted circuit is shown in Fig.9-23.

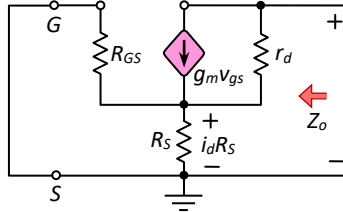


Fig.9-22: AC equivalent circuit when the input is shorted

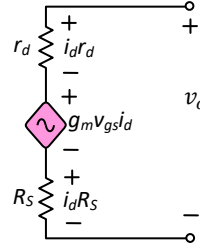


Fig.9-23: Equivalent circuit of Fig.8-21 after source conversion

Now applying KVL around this circuit [Fig.(b)] we get,

$$v_o = i_d r_d + g_m v_{gs} r_d + i_d R_S \quad (9-23)$$

$$\because v_{gs} = i_d R_S, \quad v_o = i_d (r_d + g_m R_S r_d + R_S) \quad (9-24)$$

Now, the output impedance at the drain terminal can be calculated as,

$$Z_d = \frac{v_o}{i_d} = (r_d + g_m R_S r_d + R_S) \quad (9-25)$$

The overall output impedance of the circuit will be

$$Z_o = R_D \parallel Z_d \quad (9-26)$$

But, Z_d is very large compared to R_D . Therefore,

$$Z_o \approx R_D \quad (9-27)$$

Moreover, if we calculate the output considering the load resistance R_L , it will be,

$$Z_o = R_D \parallel R_L \quad (9-28)$$

Voltage Gain:

We have already calculated the value of input voltage as [Equ.(9-16)],

$$v_i = v_{gs} (1 + g_m R_S) \quad (9-29)$$

Neglecting r_d , the output voltage will be,

$$v_o = -i_d (R_D \parallel R_L) = -g_m v_{gs} (R_D \parallel R_L) \quad (9-30)$$

Therefore, the voltage gain of the amplifier is,

$$A_v = \frac{v_o}{v_{in}} = -\frac{g_m v_{gs} (R_D \parallel R_L)}{v_{gs} (1 + g_m R_S)} = -\frac{g_m (R_D \parallel R_L)}{1 + g_m R_S} \quad (9-31)$$

$$\text{Generally, } g_m R_S \gg 1 \quad \therefore A_v = -\frac{g_m (R_D \parallel R_L)}{g_m R_S} = -\frac{(R_D \parallel R_L)}{R_S} \quad (9-32)$$

Current Gain:

Input current will be,
$$i_i = \frac{v_i}{Z_i} = \frac{v_{gs}}{R_G} \quad (9-33)$$

Output current will be (considering r_d is very large),

$$i_o = \frac{g_m v_{gs} R_D}{R_L + R_D} \quad (9-34)$$

Current gain of the amplifier will be,

$$A_i = \frac{i_o}{i_i} = \frac{g_m v_{gs} R_D}{R_L + R_D} \times \frac{R_G}{v_{gs}} = \frac{g_m R_D R_G}{R_L + R_D} \quad (9-35)$$

Example 9-4

Analyze the circuit of Example 9-2 considering the source resistor is unbypassed.

Solution:

Here, $R_G = R_1 \parallel R_2 = 4.3 \text{ M} \parallel 1 \text{ M} \approx 811 \text{ k} \text{ [Ans.]}$

and $R'_L = R_D \parallel R_L = 1.5 \text{ k} \parallel 3 \text{ k} = 1 \text{ k} \text{ [Ans.]}$

Z_i :

Using Equ.(9-), $Z_i = R_1 \parallel R_2 = 4.3 \text{ M} \parallel 1 \text{ M} \approx 811 \text{ k} \text{ [Ans.]}$

Z_o :

Using Equ.(9-20), $Z_o \approx R_D = 1.5 \text{ k} \text{ [Ans.]}$

A_v :

Using Equ.(9-32),
$$A_v = -\frac{g_m (R_D \parallel R_L)}{1 + g_m R_S} = -\frac{6 \text{ mS} \times 1 \text{ k}}{1 + 6 \text{ mS} \times 1 \text{ k}} = -0.86 \text{ [Ans.]}$$

A_i :

Using Equ.(9-35),
$$A_i = \frac{g_m R_D R_G}{R_L + R_D} = \frac{6 \text{ mS} \times 1.5 \text{ k} \times 811 \text{ k}}{3 \text{ k} + 1.5 \text{ k}} = 1.6$$

Comments: For unbypassed CS amplifier all parameters remain almost same, only the voltage gain becomes very low.

9.7 FET Common Drain Amplifier

FET common drain (CD) amplifier is similar to common collector amplifier of BJT. It **provides highest current gain, and input impedance, and lowest voltage gain and output impedance**. The voltage gain of this amplifier is approximately unity. For this reason, common drain amplifier is also called voltage follower or source follower. This type of amplifier is mainly used for impedance matching,

with very small amplitude. When this signal appears to the gate through C_1 , it superimposes with the gate biasing voltage V_G [Fig.9-26(b)]. For the positive half-cycles, the gate voltage increases and consequently drain current also increases. As the source current is equal to drain current, with the increase in input voltage, source current also increases which also increases the voltage drop across R_S . As the output is taken across R_S , the output voltage increases with the increase in input voltage []. For the negative half-cycles of output the gate voltage decreases, the source current decreases and hence, the source voltage increases. Thus, the common drain amplifier produces 0° phase difference between the input, and output signals. The output signal, developed at the drain terminal, has a DC voltage level as shown in Fig.9-26(c). But, when this signal passes through the coupling capacitor (C_2), and

appears at the load resistance, the DC voltage level is blocked and a pure AC output is produced as shown in Fig.9-26(d).

9.7.2 AC Analysis of CD Amplifier

By shorting all the capacitors, and voltage sources, the AC equivalent circuit of the CD amplifier is drawn in Fig.9-27. By replacing the FET by its equivalent model, the final AC equivalent circuit is drawn in Fig.9-28.

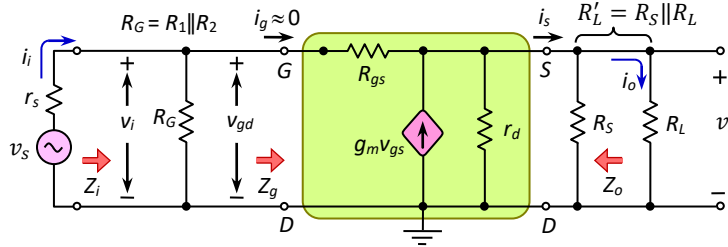


Fig.9-28: AC waveforms at different terminals of CD amplifier (with FET model)

Voltage gain:

We know that for any JFET, the gate to source current is almost zero. Only the current of the constant current-source ($g_m v_{gs}$) will flow in the output circuit. Therefore, output voltage will be,

$$v_o = g_m v_{gs} (r_d \parallel R_S \parallel R_L) \quad (9-36)$$

Looking into the input terminal, the expression for the input voltage can be written as,

$$v_i = v_{gd} = v_{gs} + v_o \quad (9-37)$$

Putting the value of v_o from Equ.(9-36) into Equ.(9-37), we get,

$$v_i = v_{gd} = v_{gs} + g_m v_{gs} (r_d \parallel R_S \parallel R_L) \quad (9-38)$$

Now, the voltage gain of the amplifier can be calculated using Equ. (9-36) and Equ.(9-38) as,

$$A_v = \frac{v_o}{v_i} = \frac{g_m v_{gs} (r_d \parallel R_S \parallel R_L)}{v_{gs} [1 + g_m (r_d \parallel R_S \parallel R_L)]} = \frac{g_m (r_d \parallel R_S \parallel R_L)}{1 + g_m (r_d \parallel R_S \parallel R_L)} \quad (9-39)$$

If $r_d \gg R_S \parallel R_L$, we can write,

$$A_v = \frac{v_o}{v_{in}} = \frac{g_m (R_S \parallel R_L)}{1 + g_m (R_S \parallel R_L)} \quad (9-40)$$

Again, if $g_m (R_S \parallel R_L) \gg 1$, the above equation reduces to,

$$A_v = 1 \quad (9-41)$$

Input Impedance:

Looking into the gate terminal, the input impedance will be,

$$Z_g = R_{gs} + r_d \parallel R_S \parallel R_L \quad (9-42)$$

Therefore, the impedance at the input terminal will be,

$$Z_i = R_G \parallel (R_{gs} + r_d \parallel R_S \parallel R_L) \quad (9-43)$$

Generally, $(R_{gs} + r_d \parallel R_S \parallel R_L)$ is much larger than R_G . Therefore, we can write,

$$Z_i \approx R_G \quad (9-44)$$

Output Impedance:

To calculate the output impedance, we have to remove the input signal that is to replace the input signal source by its internal resistance. So, source resistance r_s and bias resistance R_G will be in parallel. Now we have to apply a known amount voltage at the output terminal and the developed current for that voltage have to be calculated. From the voltage and current the output impedance will be determined. As shown in Fig.9-28, i_s amount of current produces v_o amount of voltage. So, if we apply v_o amount of voltage with the same polarity, i_s amount of current will be produced. The situation is shown in Fig.9-29. As shown in this figure, the applied voltage (v_o) will be divided between $r_s \parallel R_G$ and R_{gs} . Generally, the source resistance r_s is very small, $r_s \parallel R_G$ will also be very small. On the other hand, the gate source leakage resistance R_{gs} is very large. It will be reasonable to assume that all of the applied voltage v_o will appear across R_{gs} . Therefore,

$$v_o \approx v_{gs} \quad (9-45)$$

Due to this gate source voltage (v_{gs}), the current-source of the FET model will produce $g_m v_{gs}$ amount of current. As the current flowing through R_{gs} is almost zero, from Fig.9-28, we can write,

$$i_s \approx g_m v_{gs} \quad (9-46)$$

Looking at the constant current-source terminal (point 'a' in Fig.9-29), we find that the applied voltage is v_o and the current is i_s . Therefore, the impedance at this terminal will be,

$$Z_a = \frac{v_o}{i_s} = \frac{v_{gs}}{g_m v_{gs}} = \frac{1}{g_m} \quad (9-47)$$

Therefore, the impedance at the output terminal will be,

$$Z_o = Z_a \parallel r_d \parallel R_S = \frac{1}{g_m} \parallel r_d \parallel R_S \quad (9-48)$$

Here also, we can neglect r_d , as it is very large. Therefore,

$$Z_o = \frac{1}{g_m} \parallel R_S = \frac{1}{g_m} \parallel R_S \quad (9-49)$$

Now, we recall that the value of g_m for a small signal JFET is 1 mS to 30 mS. So, the value of Z_a will be 1 k Ω to 33 Ω . Generally, R_S is quite large than this value. Therefore, we can finally write,

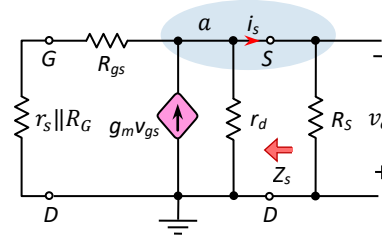


Fig.9-29: Equivalent circuit of CD with AC signal source replaced by its internal resistance

$$Z_o = \frac{1}{g_m} \text{ [if } R_S \gg 1/g_m \text{]} \quad (9-50)$$

Current Gain:

As the current flowing through R_{gs} is almost zero (i.e., $i_g \approx 0$), we can write,

$$i_i = \frac{v_i}{Z_i} = \frac{v_{gd}}{R_G} \quad (9-51)$$

Putting the value of v_{gd} from Equ.(9-38),

$$i_i = \frac{v_{gs}[1 + g_m(r_d \parallel R_S \parallel R_L)]}{R_G}$$

$$\text{As } r_d \text{ is very large,} \quad i_i = \frac{v_{gs}[1 + g_m(R_S \parallel R_L)]}{R_G} \quad (9-52)$$

At the output stage, current $g_m v_{gs}$ is divided among r_d , R_S , and R_L . The current flowing through R_L is the output current. However, as the value of r_d is very large compared to the other two resistors, we can calculate output current as,

$$\text{Neglecting } r_d, \quad i_o = \frac{g_m v_{gs} R_S}{R_L + R_S} \quad (9-53)$$

Now, the current gain of the amplifier can be calculated using Equ.(9-52) and Equ.(9-53) as,

$$A_i = \frac{i_o}{i_i} = \frac{g_m v_{gs} R_S}{R_L + R_S} \times \frac{R_G}{v_{gs}[1 + g_m(R_S \parallel R_L)]}$$

$$\text{or,} \quad A_i = \frac{i_o}{i_i} = \frac{g_m R_S R_G}{(R_L + R_S)[1 + g_m(R_S \parallel R_L)]} \quad (9-54)$$

Again, as $g_m(R_S \parallel R_L) \gg 1$,

$$A_i = \frac{i_o}{i_i} = \frac{g_m R_S R_G}{(R_L + R_S)g_m(R_S \parallel R_L)}$$

$$\text{or,} \quad A_i = \frac{i_o}{i_i} = \frac{R_S R_G}{(R_L + R_S)} \times \frac{R_S + R_L}{R_S R_L} = \frac{R_G}{R_L} \quad (9-55)$$

Example 9-5

For the common drain amplifier of Fig.9-30, determine input impedance, output impedance, voltage gain, and current gain. Assume, $r_d = 200 \text{ k}\Omega$, and $g_m = 6.6 \text{ mS}$.

Solution:

As the value of r_d is given, we will consider it. So,
Using Equ.(9-40),

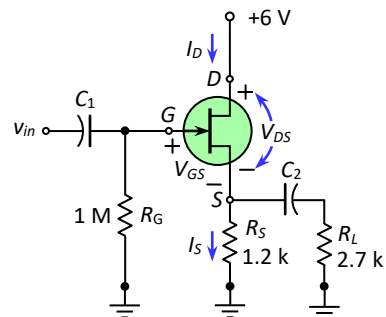


Fig.9-30: CD amplifier for Example 9-5

$$A_v = \frac{6.6 \text{ mS} \times (200 \text{ k} \parallel 1.2 \text{ k} \parallel 2.7 \text{ k})}{1 + 6.6 \text{ mS} \times (200 \text{ k} \parallel 1.2 \text{ k} \parallel 2.7 \text{ k})} \approx 0.8$$

Using Equ.(9-44),

$$Z_i \approx R_G = 1 \text{ M}$$

Using Equ.(9-49),

$$Z_o = \frac{1}{g_m} \parallel r_d \parallel R_s = \frac{1}{6.6 \text{ mS}} \parallel 200 \text{ k} \parallel 1.2 \text{ k} \approx 1 \text{ k}$$

Using Equ.(9-55),

$$A_i = \frac{R_G}{R_L} = \frac{1 \text{ M}}{2.7 \text{ k}} \approx 370$$

Comments: Like the BJT common-collector amplifier, the voltage gain of common-drain amplifier is less than unity.

9.8 Common Gate Amplifier

We have seen common gate (CG) amplifier in Chapter 8 with self-bias and voltage divider bias. The circuits are reproduced in Fig.9-31 and Fig.9-32. The input signal is applied to source terminal

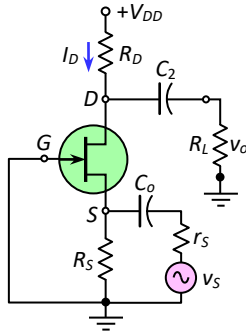


Fig.9-31: CG amplifier with self-bias

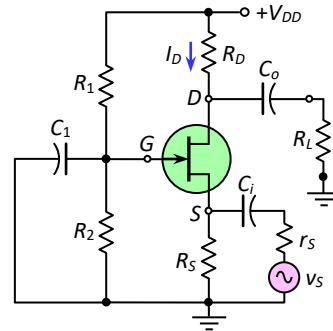


Fig.9-32: CG amplifier with voltage divider bias

through input-coupling capacitor and the load is connected to drain via the output capacitor. As the gate is common to both input and output, this circuit is called a common gate amplifier. Common gate amplifier gives high voltage gain, low input impedance and high output impedance. As mentioned in Chapter 8, the common gate amplifiers are used in circuits amplifiers that require low input impedance levels. One such application is VHF and UHF RF amplifiers where the low input impedance of the amplifier allows accurate matching to the feeder impedance which is typically 50Ω or 75Ω . Another application is moving-coil microphones preamplifiers - these microphones have very low impedance levels.

9.8.1 Operation of CG Amplifier

When the AC signal is applied to the amplifier, the current and voltage levels at different terminals of the FET changes according to the input signal. The shapes of signals at different terminals of the amplifier are shown in Fig.9-33. The input signal [Fig.9-33(a)] has both positive and negative half-cycles with very small amplitude. When this signal appears to the source through C_i , it superimposes with the source biasing voltage V_S [Fig.9-33(b)]. For the positive half-cycles, the source voltage increases and consequently V_{gs} [upper case parameter with lower case subscript

indicates combination of DC and AC signals] decrease [$\because V_{gs} = V_g - V_s$]. As we know from the transfer characteristic curve, I_d decreases with the decrease in V_{gs} . With the decrease in I_d , the voltage-drop across R_D decreases and hence, the drain voltage increases [Fig.9-33(c)]. On the other hand, for the negative half-cycles the gate voltage decreases and V_{gs} increases. With the increase in V_{gs} , the drain current increases. With the increase in I_d , the voltage drop across R_D increases, and hence the drain voltage increases. As the output is taken from drain, the output increases with the increase in input and vice versa. Thus, the common gate amplifier has no phase difference between the input, and output signals. In addition to that, due to large drain resistance (R_D), a small change in drain current produces a larger change in drain voltage. In this way, an amplified signal is developed at the output. The output signal, developed at the drain terminal, has a DC voltage level as shown in Fig.9-33(c). But, when this signal passes through the coupling capacitor (C_o), and appears across the load resistance, the DC voltage level is blocked and a pure AC output is produced as shown in Fig.9-33(d).

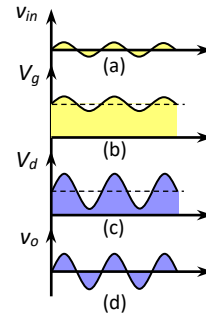


Fig.9-33: AC waveforms at different terminals of CG amplifier

9.8.2 AC Analysis of the Common Gate Amplifier

There is no gate resistance in Fig.9-31, and due to the bypass capacitor the biasing resistance (R_B) of Fig.9-32 will be shorted. Therefore, the AC equivalent circuits for both of these amplifiers will be same. Considering all the capacitors, the DC voltage supply as short circuits the AC equivalent circuit is drawn in Fig.9-34.

Finally, by replacing the JFET by its equivalent model [shown in Fig.9-9], the above circuit can be redrawn in Fig.9-35. Here, the directions of current and polarities of voltages are shown for the positive half-cycles of the input signal.

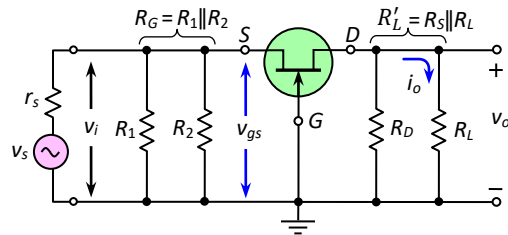


Fig.9-34: AC equivalent circuit of CG amplifier

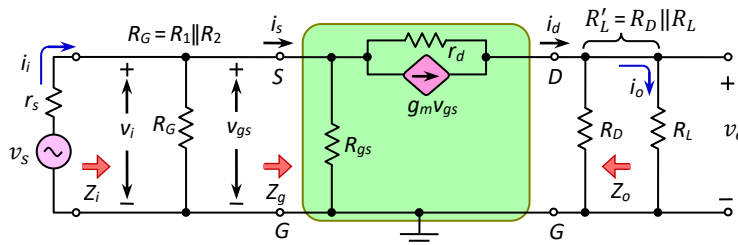


Fig.9-35: AC equivalent circuit of CG amplifier using FET model

Input Impedance:

At first, we will calculate the impedance at the source terminal. To do that let us convert the current-source ($g_m v_{gs}$) into a voltage source. Considering R_{gs} as open circuit the equivalent circuit from source terminal is shown in Fig.9-36. Now applying KVL in this circuit, we get

$$v_{gs} + g_m v_{gs} r_d - r_d i_d - i_d R'_L = 0$$

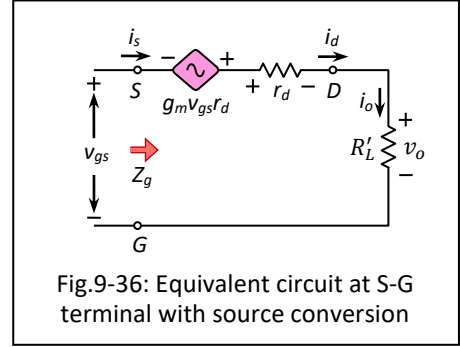
$$\therefore i_d = i_s \quad \therefore v_{gs}(1 + g_m r_d) - i_s(r_d + R'_L) = 0$$

$$\text{or,} \quad v_{gs}(1 + g_m r_d) = i_s(r_d + R'_L) \quad (9-56)$$

$$\therefore \quad \frac{v_{gs}}{i_s} = \frac{r_d + R'_L}{1 + g_m r_d} = Z_s \quad (9-57)$$

Generally, $r_d \gg R'_L$, and $g_m r_d \gg 1$. Therefore,

$$Z_s = \frac{1}{g_m} \quad (9-58)$$



Now looking into the input terminal, the overall input impedance will be,

$$Z_i = R_s \parallel Z_s \quad (9-59)$$

$$\text{or,} \quad Z_i = R_s \parallel \left\| \frac{r_d + R'_L}{1 + g_m r_d} \approx R_s \parallel \frac{1}{g_m} \right\| \quad (9-60)$$

Voltage Gain:

From the input circuit, we find that, $v_i = -v_{gs}$. Therefore, Equ.(9-56) can be written as,

$$-v_i(1 + g_m r_d) = i_s(r_d + R'_L)$$

$$\text{or,} \quad \frac{i_s}{v_i} = -\frac{1 + g_m r_d}{r_d + R'_L} \quad (9-61)$$

From the output circuit, we find that,

$$v_o = -i_d R'_L$$

$$\therefore i_d = i_s \quad \therefore v_o = -i_s R'_L \quad (9-62)$$

$$\text{or,} \quad \frac{v_o}{i_s} = -R'_L \quad (9-63)$$

Now, combining Equ.(9-61) and Equ.(9-63), we can calculate the voltage gain as,

$$A_v = \frac{v_o}{v_i} = \frac{i_s}{v_i} \times \frac{v_o}{i_s} = \left(-\frac{1 + g_m r_d}{r_d + R'_L} \right) (-R'_L) \quad (9-64)$$

As, $r_d \gg R'_L$, and $g_m r_d \gg 1$. Therefore,

$$A_v = g_m R'_L \quad (9-65)$$

$$\text{or,} \quad A_v = g_m (R_D \parallel R'_L) \quad (9-66)$$

Output Impedance:

From Equ.(9-64),
$$\frac{v_o}{v_i} = \frac{(1 + g_m r_d) R'_L}{r_d + R'_L} \quad (9-67)$$

$\therefore$
$$v_o = \frac{v_i(1 + g_m r_d) R'_L}{r_d + R'_L} \quad (9-68)$$

If we look carefully into this equation, we find that, here $v_i(1 + g_m r_d)$ amount of voltage is divided between two resistors r_d and R'_L and produces the output voltage. This equation can be represented by the circuit of Fig.9-37.

Now, to calculate the output impedance, we have to consider the voltage source (v_i) as short circuit. Therefore, the output impedance will be,

$$Z_o = r_d \parallel R_D \quad (9-69)$$

$\because r_d \gg R_D$
$$\boxed{Z_o \approx R_D} \quad (9-70)$$

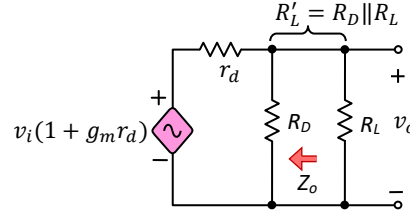


Fig.9-37: Output circuit of Fig.9-35 after source conversion

Current Gain:

We have already determined the input impedance ($Z_s = 1/g_m$) at the source terminal. The input current will be divided between Z_s and R_S . The amount current flowing through Z_s is the source current i_s . Therefore, using current divider rule, we can write,

$$i_s = \frac{i_i R_S}{R_S + 1/g_m}$$

or,
$$\frac{i_s}{i_i} = \frac{R_S}{R_S + 1/g_m} \quad (9-71)$$

Similarly, at the drain terminal, i_d is divided between R_D and R_L . The current flowing through R_L is the output current. Therefore, we can write

$$i_o = \frac{i_d R_D}{R_D + R_L}$$

or
$$\frac{i_o}{i_d} = \frac{R_D}{R_D + R_L} \quad (9-72)$$

Now, using Equ.(9-70) [considering $i_s = i_d$] and Equ.(9-71), the current gain of the amplifier can be calculated as,

$$A_i = \frac{i_o}{i_i} = \frac{i_d}{i_i} \times \frac{i_o}{i_d} = \frac{R_S}{R_S + 1/g_m} \times \frac{R_D}{R_D + R_L}$$

or,
$$\boxed{A_i = \frac{g_m R_S}{g_m R_S + 1} \times \frac{R_D}{R_D + R_L}} \quad (9-73)$$

If $g_m R_S \gg 1$, and $R_D \gg R_L$, that is, most of the current passes through the load, then

$$\boxed{A_i = \frac{g_m R_S}{g_m R_S} \times \frac{R_D}{R_D} \approx 1} \quad (9-74)$$

Example 9-6

For the common gate amplifier of Fig.9-38, determine input impedance, output impedance, voltage gain, and current gain. Assume, $r_d = 100 \text{ k}\Omega$, $V_P = -4 \text{ V}$, and $I_{DSS} = 8 \text{ mA}$.

Solution:

As g_m is not given, first of all we have to determine quiescent drain current.

The bias line of this circuit is,

$$V_{GS} = -I_D R_S = -1 \times I_D$$

Putting this value into Shockley's equation we get,

$$I_D = I_{DSS} \left(1 - \frac{1 \times I_D}{V_P} \right)^2 = 8 \text{ mA} \left(1 - \frac{1 \times I_D}{4} \right)^2$$

Solving this quadratic equation, we get $I_D = 2 \text{ mA}$.

Now, we can calculate the value of g_m using Equ.(7-12) of Chapter 7.

$$g_m = \frac{2\sqrt{I_{DSS}I_D}}{|V_P|} = \frac{2\sqrt{8 \text{ mA} \times 2 \text{ mA}}}{4 \text{ V}} = 2 \text{ mS}$$

Using Equ.(9-60),

$$Z_i = R_S \parallel \frac{1}{g_m} = 1000 \Omega \parallel \frac{1}{2 \text{ mS}}$$

$$Z_i = 510 \Omega \parallel 406.5 \Omega \approx 333 \Omega \text{ [Ans.]}$$

Using Equ.(9-66),

$$A_v = g_m R_D \parallel R_L = 2 \text{ mS} \times (5.6 \text{ k} \parallel 10 \text{ k}) \approx 7.2 \text{ [Ans.]}$$

Using Equ.(9-70),

$$Z_o \approx R_D = 5.6 \text{ k [Ans.]}$$

Using Equ.(9-73),

$$A_i = \frac{g_m R_S}{g_m R_S + 1} \times \frac{R_D}{R_D + R_L}$$

or,

$$A_i = \frac{2 \text{ mS} \times 1 \text{ k}}{2 \text{ mS} \times 1 \text{ k} + 1} \times \frac{5.6 \text{ k}}{5.6 \text{ k} + 10 \text{ k}} \approx 0.24 \text{ [Ans.]}$$

Comments: Common gate amplifier has very large voltage gain and output impedance, but very low input impedance and current gain.

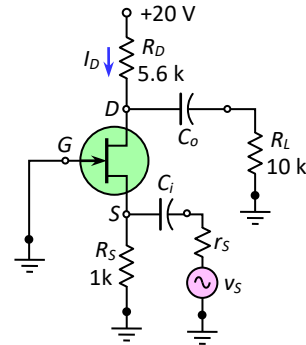


Fig.9-38: CG amplifier for Example 9-6

9.9 Comparison of CS, CD, and CG Amplifiers

So far, we have analyzed JFET amplifiers in three configurations. These are common source (CS), common drain (CD), and common gate (CG) amplifiers. We have determined voltage gain, input impedance, output impedance, and current gain of these amplifiers. We have found that the properties of CS amplifier are similar to the properties of BJT common emitter (CE) amplifier,

properties of CD amplifier are similar to the preterits of CC amplifier, and CG amplifier is similar to CB amplifier. CS amplifiers give high voltage gain, current gain, input impedance, and medium output impedance. CD amplifiers give unity voltage gain, high current gain, and input impedance, and low output impedance. CG amplifiers give high voltage gain, unity current gain, high input impedance, and low output impedance. For better comparison, all of these parameter values of three amplifiers shown in Table 8-1.

Table 8-1: Parameters of different FET amplifiers

Parameters	CS	CS (Unbypassed)	CD	CG
A_v	$\approx -g_m(R_D \parallel R_L)$	$\approx -\frac{(R_D \parallel R_L)}{R_S}$	≈ 1	$g_m(R_D \parallel R_L)$
Z_i	$\approx R_G$	$\approx R_G$	$\approx R_G$	$\approx R_S \parallel \frac{1}{g_m}$
Z_o	$\approx R_D$	$\approx R_D$	$\approx \frac{1}{g_m} \parallel R_S$	$\approx R_D$
A_i	$\approx \frac{g_m R_G R_D}{R_L + R_D}$	$\approx \frac{g_m R_G R_D}{R_L + R_D}$	$\approx \frac{R_G}{R_L}$	≈ 1

To compare the performance of FET amplifiers with BJT amplifiers the same parameters of BJT amplifiers are shown in Table 8-2.

Table 8-2: Parameters of different BJT amplifiers

Parameters	CE	CE (Unbypassed)	CC	CB
A_v	$\approx -\frac{h_{fe}(R_C \parallel R_L)}{h_{ie}}$	$\approx -\frac{h_{fe}R'_L}{h_{ie} + (h_{fe} + 1)R_E}$	$\frac{h_{fc}R'_L}{h_{ic} + h_{fc}R'_L} \approx 1$	$\frac{h_{fb}(R_C \parallel R_L)}{h_{ib}}$
Z_i	$\approx h_{ie}$	$R_B \parallel [h_{ie} + (h_{fe} + 1)R_E]$	$R_B \parallel h_{fc}(R_C \parallel R_L)$	$\approx h_{ib}$
Z_o	$\approx R_C$	$\approx R_C$	$\left(\frac{r_s \parallel R_B + h_{ic}}{h_{fc}}\right) \parallel R_E$	$\approx R_C$
A_i	$\frac{h_{fe}R_B R_C}{(R_B + Z_b)(R_C + R_L)}$	$\frac{h_{fe}R_B R_C}{(R_B + Z_b)(R_C + R_L)}$	$\approx \frac{h_{fc}R_E}{R_E + R_L}$	≈ 1

9.10 Equivalent Circuit of Depletion MOSFET

The operation and characteristics of DMOSFETs are similar to those of JFETs. The only one difference is, DMOSFETs can work in both depletion mode, and enhancement mode. That is, for N-channel and P-channel DMOSFETs, the gate can be made both positive and negative. If the gate of

an N-channel DMOSFET is biased positive, it will work in the enhancement region of its transfer characteristics. In that case, drain current will be greater than I_{DSS} . So, the value of transconductance will be greater than g_{m0} . However, in Chapter 6, we have seen that Shockley's equation is also applicable for DMOSFETs. Therefore, the value of g_m can be calculated using the equations [Equ.(6-4) and Equ.(6-5)] derived in Chapter 6. The equations are,

$$g_m = \frac{2\sqrt{I_{DSS}I_D}}{|V_P|}$$

and,

$$g_m = \frac{2I_{DSS}}{|V_P|} \left(1 - \frac{V_{GS}}{V_P}\right)$$

As the operation process and characteristics of DMOSFET are similar to those of JFET, the AC equivalent circuit of DMOSFET will be similar. In JFET, leakage current flows through the gate terminal, and for this current we introduced R_{gs} resistance in the equivalent circuit. But in MOSFET, current does not flow through the gate at all. Therefore, the small signal AC equivalent circuit of DMOSFET can be represented as,

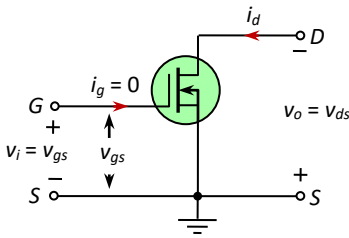


Fig.9-39: DMOSFET in CSC

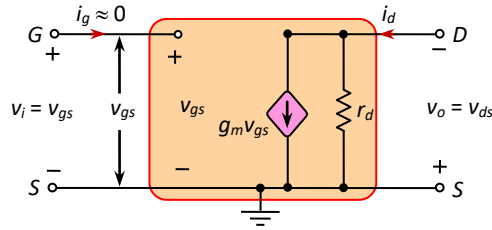


Fig.9-40: Equivalent model of DMOSFET in CSC

9.10.1 Common Source Amplifier with DMOSFET

In Chapter 8, we have discussed three types of biasing circuits for DMOSFETs in common source configuration. These are fixed bias, self-bias, and voltage divider bias. If the source resistor is properly bypassed, the AC analysis for all of them will be exactly same. In case of self-bias the gate biasing resistance will be R_G , and case of voltage divider bias the gate biasing resistance will be $R_1 \parallel R_2$. Therefore, here we will analyze only the DMOSFET CS amplifier with voltage divider biasing circuit. The circuit diagram of the amplifier is shown in Fig.9-41. The AC equivalent circuit of this amplifier is shown in Fig.9-42. All the expressions derived for the JFET CS amplifier can be used for this amplifier considering R_{gs} as open.

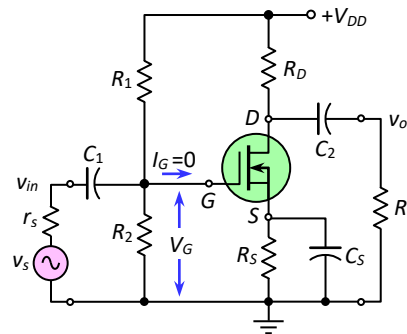


Fig.9-41: CS amplifier using DMOSFET

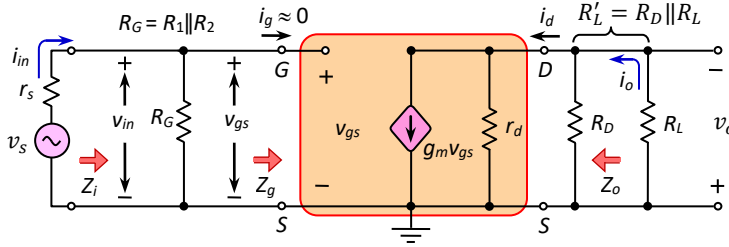


Fig.9-42: AC equivalent circuit of CS amplifier

Input Impedance:

$$Z_i = R_G = R_1 || R_2 \quad (9-75)$$

Output Impedance:

$$Z_o \approx R_D || R_L \quad (9-76)$$

Voltage Gain:

$$A_v = -g_m(R_D || R_L) = -g_m R'_L \quad (9-77)$$

Current Gain:

$$A_i = \frac{g_m R_D R_G}{R_L + R_D} \quad (9-78)$$

Example 9-7

For the common source amplifier of Fig.9-43, determine input impedance, output impedance, voltage gain, and current gain. Assume, $V_P = -4$ V, and $I_{DSS} = 10$ mA.

Solution:

DC analysis of this circuit was done in Chapter 8. The value of quiescent was 5.13 mA.

Now, we can calculate the value of g_m using Equ.(6-).

$$g_m = \frac{2\sqrt{I_{DSS}I_D}}{|V_P|} = \frac{2\sqrt{10 \text{ mA} \times 5.13 \text{ mA}}}{4 \text{ V}} = 3.58 \text{ mS}$$

Using Equ.(9-75), $Z_i = R_G = R_1 || R_2 = 4 \text{ M} || 1 \text{ M} = 800 \text{ k} [\text{Ans.}]$

Using Equ.(9-76), $Z_o \approx R_D || R_L = 2 \text{ k} || 10 \text{ k} \approx 1.67 \text{ k} [\text{Ans.}]$

Using Equ.(9-77), $A_v = -g_m R'_L = -3.58 \text{ mS} \times 1.67 \text{ k} \approx -6 [\text{Ans.}]$

Using Equ.(9-78), $A_i = \frac{g_m R_D R_G}{R_L + R_D} = \frac{3.58 \text{ mS} \times 2 \text{ k} \times 800 \text{ k}}{10 \text{ k} + 2 \text{ k}} \approx 477 [\text{Ans.}]$

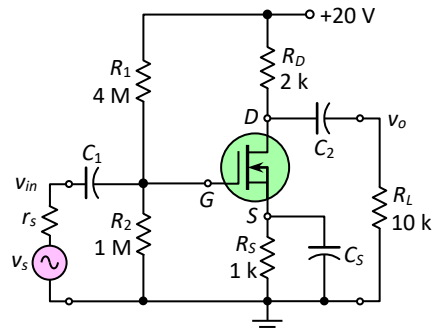


Fig.9-43: CS amplifier for Example 9-7

Comments: Calculation process is exactly same as JFET amplifiers.

9.11 Equivalent Circuit of Enhancement MOSFET

The operation and characteristics of enhancement MOSFETs are different from other FETs. The operation of EMOSFET is similar to the operation of BJT. Drain current does not flow, unless the gate-source voltage is greater than the threshold voltage (V_T). There is not V_P and I_{DSS} for EMOSFETs. Either the value of g_m or k is given for this type of FETs. The drain current is given by [equation of transfer characteristic curve],

$$I_D = k(|V_{GS}| - |V_T|)^2$$

Here, V_{GS} is the quiescent value of gate-source voltage. As described in Chapter 6, the value of g_m can be calculated using either of the following equations,

$$g_m = 2k(|V_{GS}| - |V_T|) \quad (9-79)$$

$$g_m = 2\sqrt{I_D k} \quad (9-80)$$

Although the equation of transfer characteristic curve is different from those of other two FETs, the properties of the output characteristics are almost same. With the increase of V_{DS} , I_D increases in the voltage-controlled resistance region, and then becomes almost constant in the pinch-off region. Like DMOSFET, here also there is no gate current. Therefore, the equivalent circuit of EMOSFET can also be represented as shown in Fig.9-45.

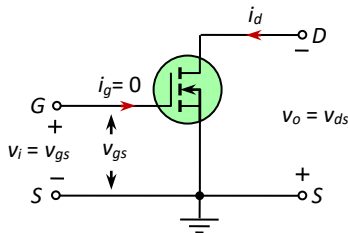


Fig.9-44: EMOSFET in CSC

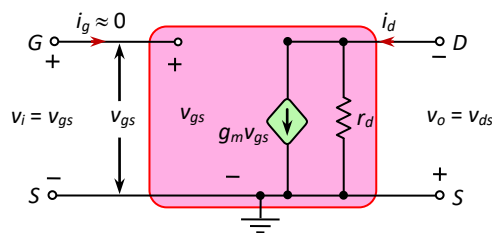


Fig.9-45: Equivalent model of EMOSFET in CSC

9.11.1 Common Source Amplifier with EMOSFET

In Chapter 8, we have discussed three types of biasing circuits for EMOSFETs in common source configuration. These are fixed bias, voltage divider bias, and **drain-feedback** bias. At first, we will consider the first two biasing circuits. Two CS amplifier circuits using EMOSFETs are shown in Fig.9-46, and Fig.9-47. The first one uses fixed bias, and second one uses the voltage divider bias. As the source resistor is bypassed by C_S , the AC equivalent circuits of these amplifiers will be exactly same. The AC equivalent circuit is shown in Fig.9-48. Here also, all the expressions derived for the JFET CS amplifier can be used considering R_{gs} as open. Therefore,

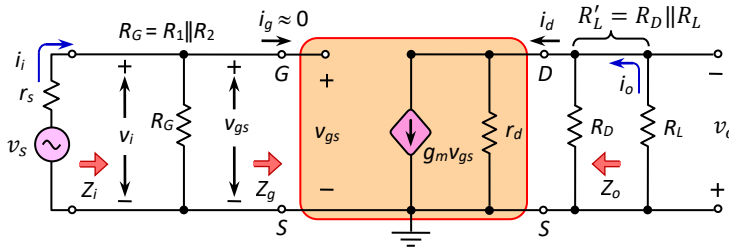


Fig.9-48: AC equivalent circuit of CS amplifier

Input Impedance:

$$Z_i = R_G = R_1 || R_2 \quad (9-81)$$

Output Impedance:

$$Z_o \approx R_D || R_L \quad (9-82)$$

Voltage Gain:

$$A_v = -g_m(R_D || R_L) = -g_m R'_L \quad (9-83)$$

Current Gain:

$$A_i = \frac{g_m R_D R_G}{R_L + R_D} \quad (9-84)$$

Example 9-8

2N7002 EMOSFET is used in the CS amplifier of Fig.9-49. This MOSFET has $V_T = 2\text{ V}$, and $k = 8\text{ mA/V}^2$. Calculate, input impedance, output impedance, voltage gain, and current gain of this amplifier.

Solution:

Here,

$$V_G = \frac{V_{DD} R_2}{R_1 + R_2} = \frac{20\text{ V} \times 1\text{ M}}{1\text{ M} + 1.6\text{ M}} \approx 7.7\text{ V}$$

The equation of bias line of this circuit is,

$$V_{GS} = V_G - I_D R_S = 7.7\text{ V} - 1\text{ k} \times I_D$$

Putting this value into the equation of transfer characteristic curve of EMOSFET, we get,

$$I_D = k(V_{GS} - V_T)^2$$

$$\text{or, } I_D = 8\text{ mA/V}^2 (7.7\text{ V} - 1\text{ k} \times I_D - 2\text{ V})^2$$

Solving this quadratic equation, we will get,

$$I_D = 4.92\text{ mA}$$

Now, using Equ.(7-29) of Chapter 7,

$$g_m = 2\sqrt{I_D k} = 2\sqrt{4.92\text{ mA} \times 8\text{ mA/V}^2} \approx 6.27\text{ mS}$$

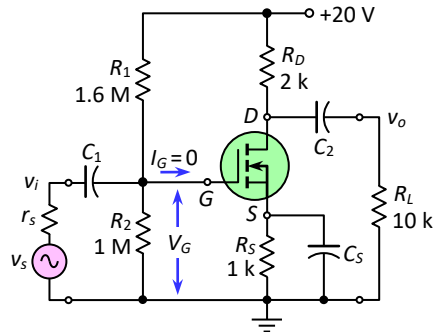


Fig.9-49: EMOSFET amplifier circuit for Example 9-8

Using Equ.(9-81), $Z_i = R_G = R_1 \parallel R_2 = 1.6 \text{ M} \parallel 1 \text{ M} \approx 615 \text{ k} [\text{Ans.}]$

Using Equ.(9-82), $Z_o \approx R_D \parallel R_L = 2 \text{ k} \parallel 10 \text{ k} \approx 1.67 \text{ k} [\text{Ans.}]$

Using Equ.(9-83), $A_v = -g_m R'_L = -6.27 \text{ mS} \times 1.67 \text{ k} \approx 10.5 [\text{Ans.}]$

Using Equ.(9-84), $A_i = \frac{g_m R_D R_G}{R_L + R_D} = \frac{6.27 \text{ mS} \times 2 \text{ k} \times 615 \text{ k}}{10 \text{ k} + 2 \text{ k}} \approx 643 [\text{Ans.}]$

Comments: Analysis of EMOSFET amplifier is same as that of JFET and DMOSFET amplifiers.

9.11.2 EMOSFET Amplifier with Drain-Feedback Bias

Drain-feedback bias is similar to the collector-feedback bias of BJT. This type of biasing circuit can only be used with EMOSFETs, because the operation of EMOSFET is similar to the operation of BJT. A common source EMOSFET amplifier with drain-feedback biasing is shown in Fig.9-50. DC analysis of this circuit has been covered in Chapter 8.

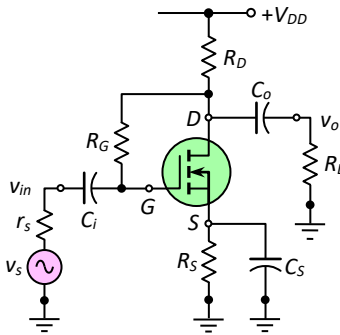


Fig.9-50: EMOSFET amplifier with drain-feedback bias

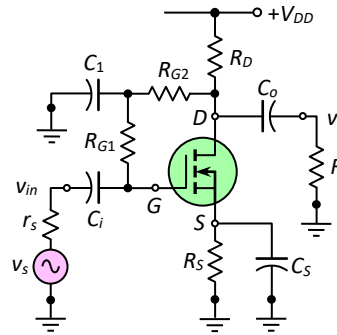


Fig.9-51: Drain-feedback bias with split gate resistance

Before going to AC analysis, we will discuss a degenerative effect that occurs for the AC signal in this circuit. The gate voltage of this circuit is,

$$V_{GS} = V_{DD} - I_D(R_D + R_S) \quad (9-85)$$

Therefore, if the drain current increases, the gate voltage will decrease. So the drain current will be adjusted back to the initial value. Similarly, if the drain current decreases, according to the above equation, the gate-source voltage will increase, and the drain current will be adjusted again. This adjusting property of this circuit is called *negative feedback* and gives a better stability factor for the operating point. However, for the AC signal this process is not desired. For example, when the AC input signal increases in positive direction, the drain current will increase, which is essential to get amplified

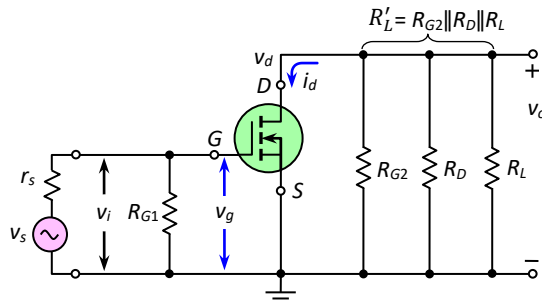


Fig.9-52: AC equivalent circuit of EMOSFET amplifier with drain-feedback bias

output signal. But the negative feedback tends to decrease the gate voltage and reduces the change in drain current. In this way, this negative feedback decreases the gain of the amplifier. This is called **degenerative effect**. Degenerative effect was also described for BJT with drain-feedback bias. The solution is exactly same as the BJT circuits. We have to split the gate resistance (R_G) into two equal resistances ($R_G = R_{G1} + R_{G2}$), and a capacitor has to be connected to the common points of the resistors and ground as shown in Fig.9-51. Now, the circuit will provide negative feedback for DC condition, but as the AC signal will not appear to the gate, negative feedback will not be effective for the AC signal.

To draw the AC equivalent circuit of this amplifier we have to consider all capacitors and DC voltage sources as short. The AC equivalent circuit is shown in Fig.9-52. This circuit is comparable to the AC equivalent circuits of other EMOSFET amplifier. The only difference is, now the effective gate resistance will be R_{G1} , and the effective AC load resistance will be $R_{G2} || R_D || R_L$. Using these changes all of the previously derived equations can be used. The process is illustrated in Example 9-9.

Input Impedance:

$$Z_i = R_{G1} \quad (9-86)$$

Output Impedance:

$$(Z_o \approx R_{G2} || R_D || R_L) \quad (9-87)$$

Voltage Gain:

$$A_v = -g_m(R_{G2} || R_D || R_L) = -g_m R'_L \quad (9-88)$$

Current Gain:

$$A_i = \frac{g_m R_{G1} (R_D || R_{G2})}{(R_D || R_{G2}) + R_L} \quad (9-89)$$

Example 9-9

For the CS amplifier of Fig.9-53, determine input impedance, output impedance, voltage gain, and current gain. Assume, $V_T = 3$ V, and $g_m = 2.2$ mS.

Solution:

For this circuit $R_G = R_{G1} = 510$ k, and

$R'_L = R_{G2} || R_D || R_L = 510 \text{ k} || 1.5 \text{ k} || 10 \text{ k} \approx 1.3 \text{ k}$

Using Equ.(9-86),

$$Z_i = R_G = R_{G1} = 510 \text{ k [Ans.]}$$

Using Equ.(9-87),

$$Z_o \approx R_{G2} || R_D || R_L \approx 1.3 \text{ k [Ans.]}$$

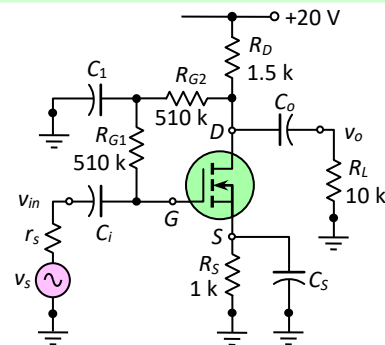


Fig.9-53: EMOSFET amplifier circuit for Example 9-9

Using Equ.(9-88), $A_v = -g_m R'_L = -2.2 \text{ mS} \times 1.3 \text{ k} = 2.86 \text{ [Ans.]}$

Comments: Calculation process is exactly same as other CS amplifiers.

9.12 High Frequency Model of FETs

The equivalent circuits of FETs used so far, are low frequency small signal models. These models can be used for low and mid-band frequencies. Like BJT, at high frequencies, we have to consider the parasitic capacitances of FETs. In Chapter 6, we have discussed these capacitances.

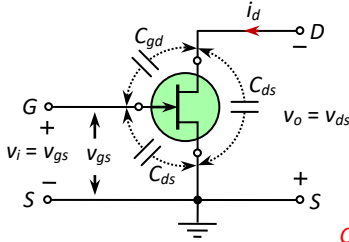


Fig.9-54: Different parasitic capacitances of JFET

Consider C_w when used in circuit

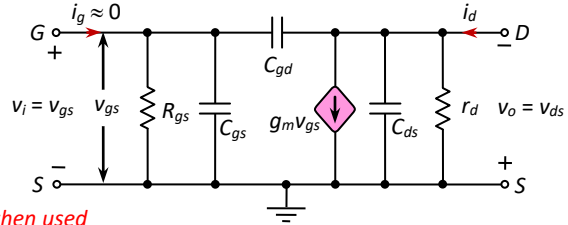


Fig.9-55: Equivalent circuit considering parasitic capacitances

There are mainly three inter-electrode capacitances. These are **gate-drain capacitance** (C_{gd}), **gate-source capacitance** (C_{gs}), and **drain-source capacitance** (C_{ds}) as shown in Fig.9-54. To draw the high-frequency equivalent circuit of FETs, we have to consider these capacitances in the equivalent of circuit of Fig.9-6. C_{gs} will come in parallel with R_{gs} , and C_{ds} will come in parallel with r_d . The drain-source capacitance (C_{ds}) will be connected between input (gate) and output (drain) terminals (Fig.9-54). Therefore, the high-frequency equivalent model of FETs will be like the circuit of Fig.9-55.

As, C_{gd} is connected between input and output, it will be difficult to analyze this circuit. But, as described in Chapter 5, using Miller's theorem, this capacitance can be split into two capacitances: one in the input side (C_{Mi}), and the other in the output side (C_{Mo}). The common-source amplifier produces 180° phase shift, and the voltage gain of the device is $g_m r_d$. Therefore, using the Miller's theorem, the value of C_{Mi} , and C_{Mo} can be calculated as,

$$C_{Mi} = C_{gd}(1 + g_m r_d) \quad (9-90)$$

and,

$$C_{Mo} = \frac{C_{gd}(1 + g_m r_d)}{g_m r_d} \quad (9-91)$$

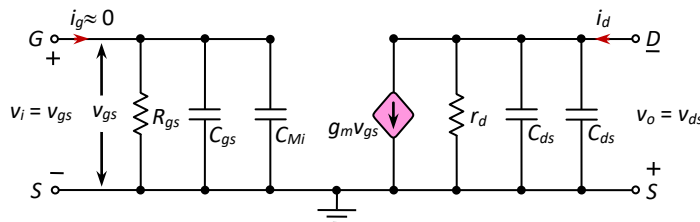


Fig.9-56: C_{gd} is split using Mitter's theorem

Now, the equivalent circuit of Fig.9-55 can be redrawn as in Fig.9-56. Finally, combining the input capacitances and the output capacitances the high-frequency equivalent circuit of FETs is drawn in Fig.9-57. Here, C_i is the equivalent capacitance in the input side and C_o is the equivalent capacitance in the output side. The values of these capacitances will be,

$$C_i = C_{gs} + C_{Mi} \quad (9-92)$$

$$\text{and,} \quad C_o = C_{ds} + C_{Mo} \quad (9-93)$$

In addition to these capacitances, sometimes we consider two wiring capacitances: one in the input side (C_{wi}), and the other in the output side (C_{wo}).

Although, the high frequency model of Fig.9-57 uses the inter-electrode capacitances, but the datasheets of FETs generally mention the values of C_{iss} , C_{oss} , and C_{rss} . For example, the high-frequency JFETs 2N5484-2N5486 have $C_{iss} = 5$ pF, $C_{oss} = 2$ pF, and $C_{rss} = 1$ pF. Actually, these capacitances are related to the **y-parameter equivalent circuit** of FTE. The y-parameter equivalent circuit of JFET in common source configuration is shown in Fig.9-58.

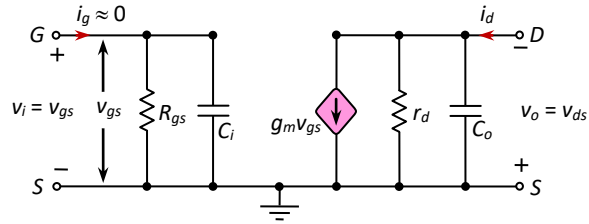


Fig.9-57: Combined all capacitances in input and output sides

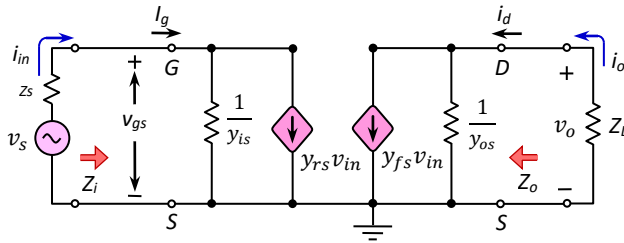


Fig.9-58: Equivalent circuit of FET amplifier using admittance parameters or y-parameters

The names and definitions of the y-parameters used in this equivalent circuit are given in Table. 8.1

Symbol	Definition	Name
y_{is}	$\left. \frac{i_g}{v_{gs}} \right _{v_{ds}=0} = \left. \frac{\Delta I_G}{\Delta V_{GS}} \right _{V_{DS}=\text{constant}}$	Common-source short-circuit input admittance
y_{rs}	$\left. \frac{i_g}{v_{ds}} \right _{v_{gs}=0} = \left. \frac{\Delta I_G}{\Delta V_{DS}} \right _{V_{GS}=\text{constant}}$	Common-source short-circuit reverse transfer admittance
y_{fs}	$\left. \frac{i_d}{v_{gs}} \right _{v_{ds}=0} = \left. \frac{\Delta I_D}{\Delta V_{GS}} \right _{V_{DS}=\text{constant}}$	Common-source short-circuit forward transfer admittance
y_{os}	$\left. \frac{i_d}{v_{ds}} \right _{v_{gs}=0} = \left. \frac{\Delta I_D}{\Delta V_{DS}} \right _{V_{GS}=\text{constant}}$	Common-source short-circuit output admittance

By neglecting the very large gate-to-source resistance (R_{gs}) of a FET, as we have done so far, it can be shown that [derivation is not shown here],

$$y_{is} = j\omega(C_{gs} + C_{gd}) \quad (9-94)$$

$$y_{fs} = g_m - j\omega C_{gd} \quad (9-95)$$

$$y_{rs} = -j\omega C_{gd} \quad (9-96)$$

$$y_{os} = 1/r_d + j\omega(C_{gd} + C_{ds}) \quad (9-97)$$

The relations of the parameters mentioned in datasheets with the y -parameters are given below.

$$y_{is} = j\omega C_{iss} \quad (9-98)$$

$$y_{fs} = g_m - j\omega C_{fss} \quad (9-99)$$

$$y_{rs} = -j\omega C_{rss} \quad (9-100)$$

$$y_{os} = \frac{1}{r_d} + j\omega C_{oss} \quad (9-101)$$

Comparing Equ.(9-94) through Equ.(9-97) to Equ.(9-98) through Equ.(9-101), we can write that,

$$C_{iss} = C_{gs} + C_{gd} \quad (9-102)$$

$$C_{fss} = C_{gd} \quad (9-103)$$

$$C_{rss} = C_{gd} \quad (9-104)$$

$$C_{oss} = C_{gd} + C_{ds} \quad (9-105)$$

Now, using these equations and the values of C_{iss} , C_{oss} , and C_{rss} from the datasheets, the values of inter-electrode capacitances of FETs can easily be calculated.

Example 9-10

The radio frequency switching JFETs 2N5484-2N5486 have $C_{iss} = 5$ pF, $C_{oss} = 2$ pF, and $C_{rss} = 1$ pF. Calculate the inter-electrode capacitance of these FETs.

Solution:

Using Equ.(9-104), $C_{gd} = C_{rss} = 1$ pF [Ans.]

Using Equ.(9-102), $C_{gs} = C_{iss} - C_{gd} = 5$ pF $- 1$ pF = 4 pF [Ans.]

Using Equ.(9-105), $C_{ds} = C_{oss} - C_{gd} = 2$ pF $- 1$ pF = 1 pF [Ans.]

Comments: Parasitic resistances are very small and they become effective at very high frequency.

9.13 Amplifiers with P-channel FETs

So far, we have discussed the process of analyzing different FET amplifiers. All of them have been constructed using N-channel FETs. In Chapter 8, the different biasing techniques of P-channel FETs have been discussed. The AC analysis of these amplifiers will be exactly similar to those of N-channel FETs. Without any confusion, all of the equations derived for N-channel FET circuits can be used for the P-channel FET circuits. One example is given below.

Example 9-11

Calculate, input impedance, output impedance, voltage gain, and current gain of the amplifier of Fig.9-59. Assume, $g_m = 6 \text{ mS}$.

Solution:

Here, $R_G = R_1 \parallel R_2 = 2.2 \text{ M} \parallel 1.1 \text{ M} \approx 733 \text{ k}$

Input Impedance Z_i : Using Equ.(9-4),

$$Z_i = R_1 \parallel R_2 = 2.2 \text{ M} \parallel 1.1 \text{ M} \approx 733 \text{ k [Ans.]}$$

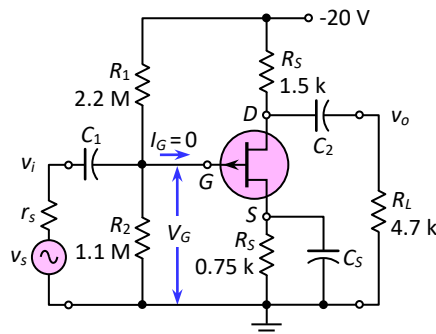


Fig.9-59: P-channel JFET amplifier for Example 9-11

Output Impedance Z_o :

Using Equ.(9-6), $Z_o \approx R_D = 1.5 \text{ k [Ans.]}$

Voltage gain A_v :

Using Equ.(9-11), $A_v = -g_m(R_D \parallel R_L) = -6 \text{ mS} \times (1.5 \text{ k} \parallel 4.7 \text{ k}) = -6.8 \text{ [Ans.]}$

Current gain A_i :

Using Equ.(9-15), $A_i = \frac{g_m R_D R_G}{R_L + R_D} = \frac{6 \times 1.5 \text{ k} \times 733 \text{ k}}{4.7 \text{ k} + 1.5 \text{ k}} = 1064$

Comments: Analysis of P-channel FET amplifier is exactly same as N-channel FET amplifiers.

9.14 Sample Questions

1. What is AC load line and how it differs from the DC load line.
2. Explain how to draw the AC load line of a FET circuit.
3. Why we draw the AC load line through the DC operating point?
4. Why we need the equivalent circuit of a device?
5. Draw the equivalent circuit of a JFET in common source configuration (CSC) and explain different parameters.
6. Draw the equivalent circuit of a JFET in common drain configuration (CDC) and explain.
7. Draw the equivalent circuit of a JFET in common gate configuration (CGC) and explain.
8. Draw the schematic diagram of a common source JFET amplifier and describe its operation. Which configuration of BJT is equivalent to this JFET amplifier?
9. Draw the JFET CS amplifier and its equivalent circuit. Derive the expression for the input impedance (Z_i), output impedance (Z_o), voltage gain (A_v), and current gain (A_i).
10. Draw the JFET CSC amplifier without the source bypass capacitor and its equivalent circuit. Derive the expression for the input impedance (Z_i), output impedance (Z_o), voltage gain (A_v), and current gain (A_i).
11. Draw the JFET CD amplifier and describe its operation with waveforms at different points of the circuit.
12. Draw the JFET CD amplifier and its equivalent circuit. Derive the expression for the input impedance (Z_i), output impedance (Z_o), voltage gain (A_v), and current gain (A_i).
13. Draw the JFET common gate (CG) amplifier and describe its operation with waveforms at different points of the circuit.
14. Draw the JFET CG amplifier and its equivalent circuit. Derive the expression for the input impedance (Z_i), output impedance (Z_o), voltage gain (A_v), and current gain (A_i).
15. Give a comparison among common source, common drain, and common gate JFET amplifiers.
16. Draw the equivalent circuit of depletion MOSFET and explain.
17. Draw the CS amplifier with depletion MOSFET and its equivalent circuit. Derive the expression for the input impedance (Z_i), output impedance (Z_o), voltage gain (A_v), and current gain (A_i).
18. Draw the equivalent circuit of enhancement MOSFET and explain.
19. Draw the CS amplifier with EMOSFET and its equivalent circuit. Derive the expression for the input impedance (Z_i), output impedance (Z_o), voltage gain (A_v), and current gain (A_i).
20. Draw the drain feedback bias circuit for EMOSFET. What are the advantages of this biasing circuit? What is degenerative effect of this biasing circuit- explain. How this problem can be overcome?
21. Describe different inter electrode capacitances of JFET.
22. Draw the high frequency equivalent circuit for JFET and simplify it step by step using Miller theorem.
23. Draw the equivalent circuit of a FET using y-parameters. Write the names of four y-parameters used in this equivalent circuit.

9.15 Exercises

- For the common source amplifier with voltage divider network of Fig.9-3, draw the DC and AC load line. Assume, $V_P = -5\text{ V}$, and $I_{DSS} = 10\text{ mA}$.

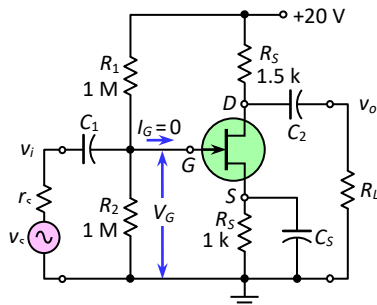


Fig.9-3: Circuit for Example 9-1

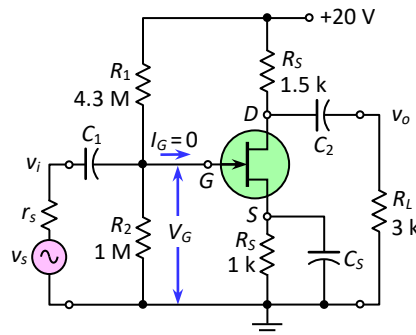


Fig.9-14: Circuit for Example 9-2

- Calculate, input impedance, output impedance, voltage gain, and current gain of the amplifier of Fig.9-14. Assume, $g_m = 6\text{ mS}$.
- For the following [Fig.9-18] common source amplifier with fixed bias, calculate input impedance, output impedance, voltage gain, and current gain. Assume $g_m = 3.5\text{ mS}$.

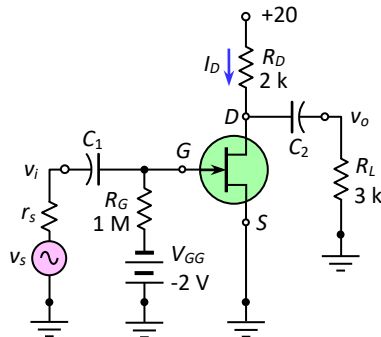


Fig.9-18: CS amplifier with fixed bias for Example 9-3

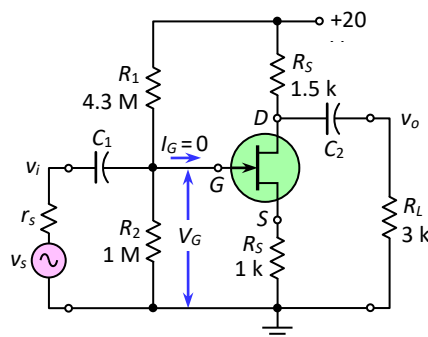


Fig.9-14: Circuit for Example 9-2

- Analyze the circuit of Example 9-2 considering the source resistor is unbypassed.
- For the common drain amplifier of Fig.9-30, determine input impedance, output impedance, voltage gain, and current gain. Assume, $r_d = 200\text{ k}\Omega$, and $g_m = 6.6\text{ mS}$.

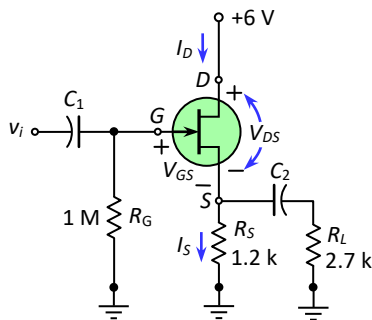


Fig.9-30: CD amplifier for Example 9-5

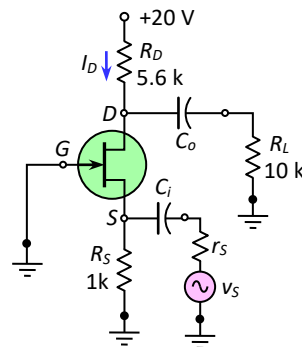


Fig.9-38: CG amplifier for Example 9-6

6. For the common gate amplifier of Fig.9-38, determine input impedance, output impedance, voltage gain, and current gain. Assume, $r_d = 100 \text{ k}\Omega$, $V_P = -4 \text{ V}$, and $I_{DSS} = 8 \text{ mA}$.
7. A For the common source amplifier of Fig.9-43, determine input impedance, output impedance, voltage gain, and current gain. Assume, $V_P = -4 \text{ V}$, and $I_{DSS} = 10 \text{ mA}$.
8. 2N7002 EMOSFET is used in the CS amplifier of Fig.9-49. This MOSFET has $V_T = 2 \text{ V}$, and $k = 8 \text{ mA/V}^2$. Calculate, input impedance, output impedance, voltage gain, and current gain of this amplifier.

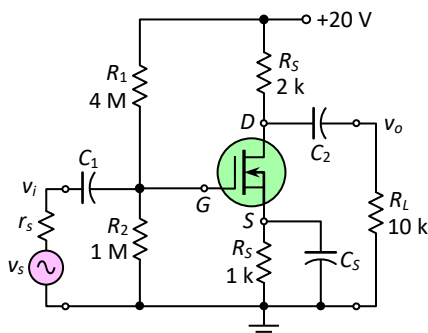


Fig.9-43: CS amplifier for Example 9-7

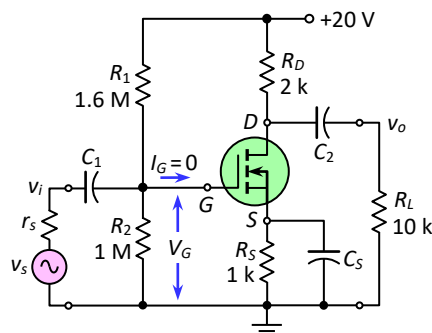


Fig.9-49: EMOSFET amplifier circuit for Example 9-8

9. For the CS amplifier of Fig.9-53, determine input impedance, output impedance, voltage gain, and current gain. Assume, $V_T = 3 \text{ V}$, and $g_m = 2.2 \text{ mS}$.
10. The radio frequency switching JFETs 2N5484-2N5486 have $C_{iss} = 5 \text{ pF}$, $C_{oss} = 2 \text{ pF}$, and $C_{rss} = 1 \text{ pF}$. Calculate the inter-electrode capacitance of these FETs.
- 11.

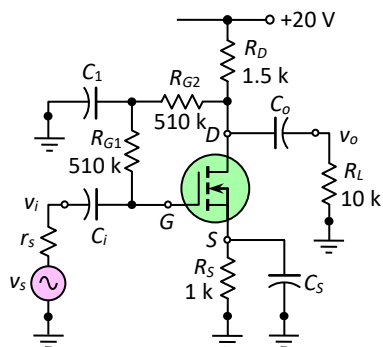


Fig.9-53: EMOSFET amplifier circuit for Example 9-9

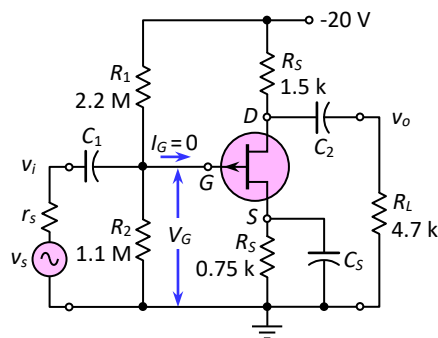


Fig.9-59: P-channel JFET amplifier for Example 9-11

12. A Calculate, input impedance, output impedance, voltage gain, and current gain of the amplifier of Fig.9-59. Assume, $g_m = 6 \text{ mS}$.